Technical Note

PlatoSim Reference Frames

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Document History

Issue	Rev.	Date	Change description
1	0	2015-07-13	Initial version.
1	1	2015-08-27	Corrected gnomonic projection. Extended Section 3.
1	2	2015-09-01	Sign corrections, so that all rotations are now clockwise.
1	3	2016-02-11	Wrote the section about yaw, pitch, and roll.
2	1	2016-02-18	New relation between spacecraft and focal plane reference frames.
3	1	2016-04-17	Adapted spacecraft reference system to the one in AD-1.
4	1	2016-05-24	Added a small section the telescope drift rotation angles.
5	1	2016-06-22	Adapted the focal plane reference frame to the one of AD-2.
6	1	2017-01-17	Homogenised w.r.t. reference frames in updated AD-2.
6	2	2017-05-29	Changed CCD (A,B,C,D) to (1,2,3,4) nomenclature.
6	3	2017-09-18	Corrected values for the orientation angle for CCDs 2 and 4.

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1 Abstract

2 Documents and Acronyms

2.1 Applicable Documents

AD-1: PLATO-OHB-PL-LI-O09, *PLATO Instrument Coordinate Systems*

AD-2: PLATO-DLR-PL-TN-016, *Geometric Camera Model for PLATO TOU*

2.2 Reference Documents

RD-1: P. Marcos-Arenal et al., 2014, A&A 566, A92, *The PLATO Simulator: Modelling of High-Cadence Space-Based Imaging*

2.3 Acronyms

CCD	Charge-Coupled Device
EQ	Equatorial reference frame
FP	Focal Plane
NP	Celestial Equatorial North Pole
PSF	Point Spread Function
SC	Spacecraft
SVM	Service Module

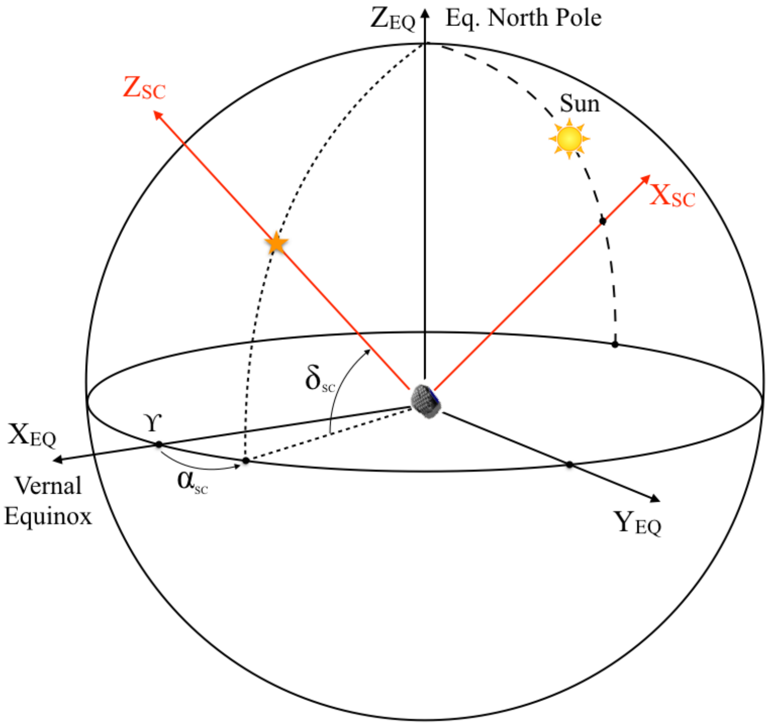

Figure 1: The base celestial equatorial coordinate system $(X_{\text{EQ}}, Y_{\text{EQ}}, Z_{\text{EQ}})$. The positive Z_{SC} axis points towards the operator-given platform pointing coordinates $(\alpha_{\text{SC}}, \delta_{\text{SC}})$.

3 The Spacecraft reference frame

The base reference system of PlatoSim is the standard celestial equatorial coordinate system $(X_{\text{EQ}}, Y_{\text{EQ}}, Z_{\text{EQ}})$ where X_{EQ} points towards the vernal equinox, and Z_{EQ} to the equatorial North Pole, as shown in Fig. 1.

The right-handed cartesian coordinate system $(X_{\text{SC}}, Y_{\text{SC}}, Z_{\text{SC}})$, associated with the spacecraft, is the same as the one defined in (AD-1). Its origin is the geometric center of the interface between the bottom of the optical bench and the service module. The positive roll axis Z_{SC} points towards the operator-given mean payload line-of-sight, given by the equatorial sky coordinates $(\alpha_{\text{SC}}, \delta_{\text{SC}})$. The coordinates of the unit vector $\hat{\mathbf{z}}_{\text{SC}}$ corresponding with the principal axis Z_{SC} can be computed in the equatorial reference frame with

$$\begin{aligned} \hat{\mathbf{z}}_{\text{SC}} &= f(1, \pi/2 - \delta_{\text{SC}}, \alpha_{\text{SC}}) \\ &= (\cos\delta_{\text{SC}}\cos\alpha_{\text{SC}}, \cos\delta_{\text{SC}}\sin\alpha_{\text{SC}}, \sin\delta_{\text{SC}})^t \end{aligned} \tag{1}$$

where f is the standard spherical coordinate transformation (22). The X_{SC} axis points in the direction of the highest point of the sunshield, as shown in Fig. 2. The spacecraft is rotated around its Z_{SC} axis, such that the X_{SC} axis intersects the great circle going through the celestial poles and the Sun. The coordinates of the corresponding unit vector $\hat{\mathbf{x}}_{\text{SC}}$ in the equatorial reference frame are

$$\hat{\mathbf{x}}_{\text{SC}} = (\cos\delta_x\cos\alpha_\odot, \cos\delta_x\sin\alpha_\odot, \sin\delta_x)^t \tag{2}$$

where $(\alpha_\odot, \delta_\odot)$ are the (mean) celestial coordinates of the Sun, and where the orthogonality requirement $\hat{\mathbf{x}}_{\text{SC}} \cdot \hat{\mathbf{z}}_{\text{SC}} = 0$ leads to the following implicit definition of δ_x:

$$\tan\delta_x = -\cot\delta_{\text{SC}}\cos(\alpha_{\text{SC}} - \alpha_\odot) \tag{3}$$

The unit vector $\hat{\mathbf{y}}_{\text{SC}}$ completes the orthonormal reference frame:

$$\hat{\mathbf{y}}_{\text{SC}} = \hat{\mathbf{z}}_{\text{SC}} \times \hat{\mathbf{x}}_{\text{SC}} \tag{4}$$

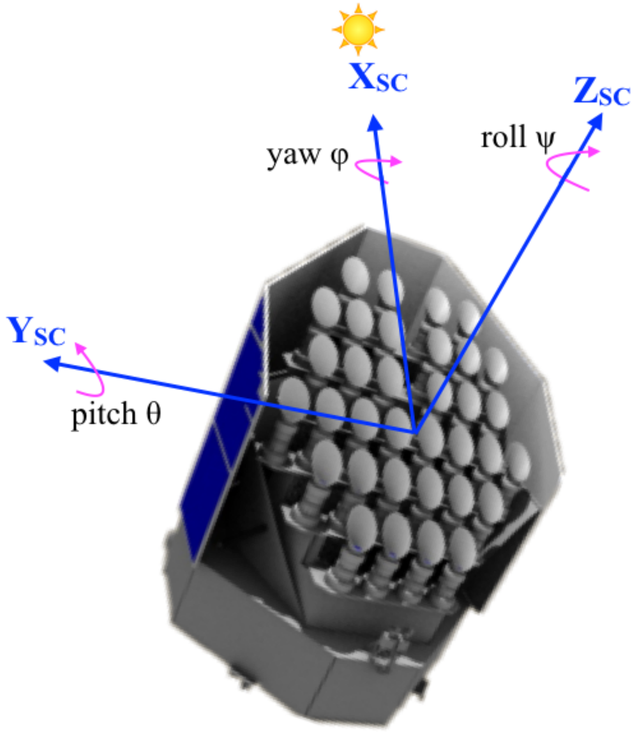

Figure 2: Definition of the right-handed $(X_{\text{SC}}, Y_{\text{SC}}, Z_{\text{SC}})$ coordinate system. The positive Z_{SC} axis points towards the operator-given pointing coordinates $(\alpha_{\text{SC}}, \delta_{\text{SC}})$. The X_{SC} axis points in the direction of the highest point of the sunshield.

Given a target with celestial equatorial sky coordinates (α, δ), its cartesian coordinates $(v_x, v_y, v_z)_{\text{SC}}$ in the spacecraft reference frame are computed using the following transformation:

$$\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix}_{\text{SC}} = \begin{pmatrix} \hat{\mathbf{x}}_{\text{SC},x} & \hat{\mathbf{x}}_{\text{SC},y} & \hat{\mathbf{x}}_{\text{SC},z} \\ \hat{\mathbf{y}}_{\text{SC},x} & \hat{\mathbf{y}}_{\text{SC},y} & \hat{\mathbf{y}}_{\text{SC},z} \\ \hat{\mathbf{z}}_{\text{SC},x} & \hat{\mathbf{z}}_{\text{SC},y} & \hat{\mathbf{z}}_{\text{SC},z} \end{pmatrix} \begin{pmatrix} \cos\delta\cos\alpha \\ \cos\delta\sin\alpha \\ \sin\delta \end{pmatrix}_{\text{EQ}} \tag{5}$$

where the equatorial cartesian coordinates of the unit vectors $\hat{\mathbf{x}}_{\text{SC}}$, $\hat{\mathbf{y}}_{\text{SC}}$, and $\hat{\mathbf{z}}_{\text{SC}}$ are given by Eqs. (1), (2), and (4).

4 The telescope reference frame

For historical reasons, throughout its code PlatoSim uses a somewhat different definition of "telescope" than used elsewhere in the Plato consortium. Telescope here is definined as the combined unit of baffle, optical elements, and detector. Elsewhere this would be called the "camera".

The right-handed cartesian coordinate system $(X_{\text{TL}}, Y_{\text{TL}}, Z_{\text{TL}})$ is associated with one of the telescopes and has its origin exactly in the middle of the 4 CCDs in the associated focal plane. The Z_{TL} axis is pointing towards the field-of-view and is called the optical axis, which is usually not parallell with the spacecraft pointing axis Z_{SC} as shown in Fig. 3. To specify how the telescope optical axis is oriented with respect to the platform, two (user-given) angles are used: the tilt angle ϱ_{TL} between Z_{TL} and Z_{SC}, and the position angle η_{TL} of a rotation around the positive Z_{SC} axis. The orientation of the unit vector $\hat{\mathbf{z}}_{\text{TL}}$ corresponding to the optical axis Z_{TL}, can be computed by first rotating the $(X_{\text{SC}}, Y_{\text{SC}})$ plane around the pointing axis Z_{SC} over an angle η_{TL} to get the reference frame $(X'_{\text{SC}}, Y'_{\text{SC}}, Z'_{\text{SC}})$, and then rotating the $Z'_{\text{SC}} = Z_{\text{SC}}$ axis over an angle ρ_{TL} around the Y'_{SC} axis. These rotations are illustrated in Fig. 4. A vector with coordinates $(x_{\text{SC}}, y_{\text{SC}}, z_{\text{SC}})$ in the spacecraft reference

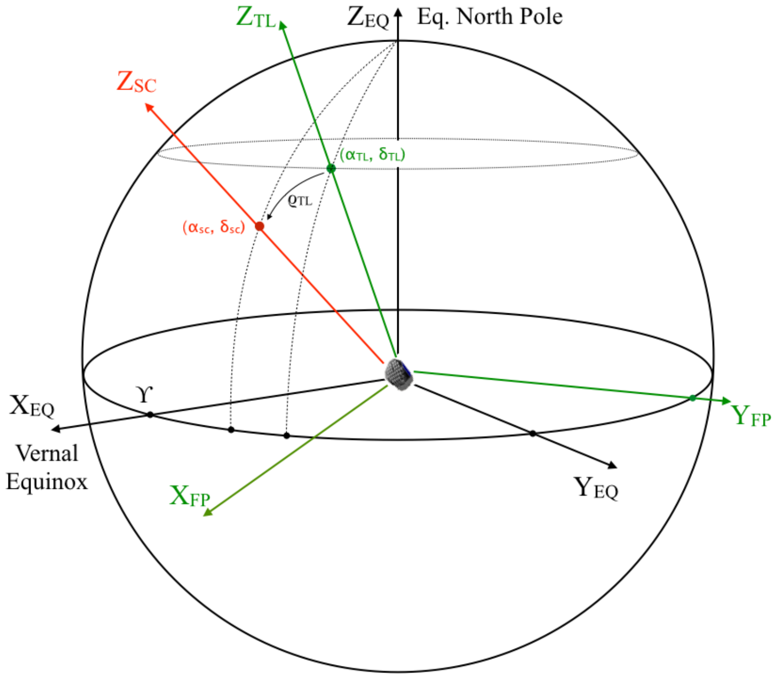

Figure 3: Definition of the Focal Plane reference frame $(X_{\rm FP}, Y_{\rm FP}, Z_{\rm FP})$. The focal plane is defined by the plane $(X_{\rm FP}, Y_{\rm FP})$, where the axis $Y_{\rm FP}$ is in the equatorial plane. $Z_{\rm FP}$ is the optical axis of the telescope, pointing towards the field-of-view.

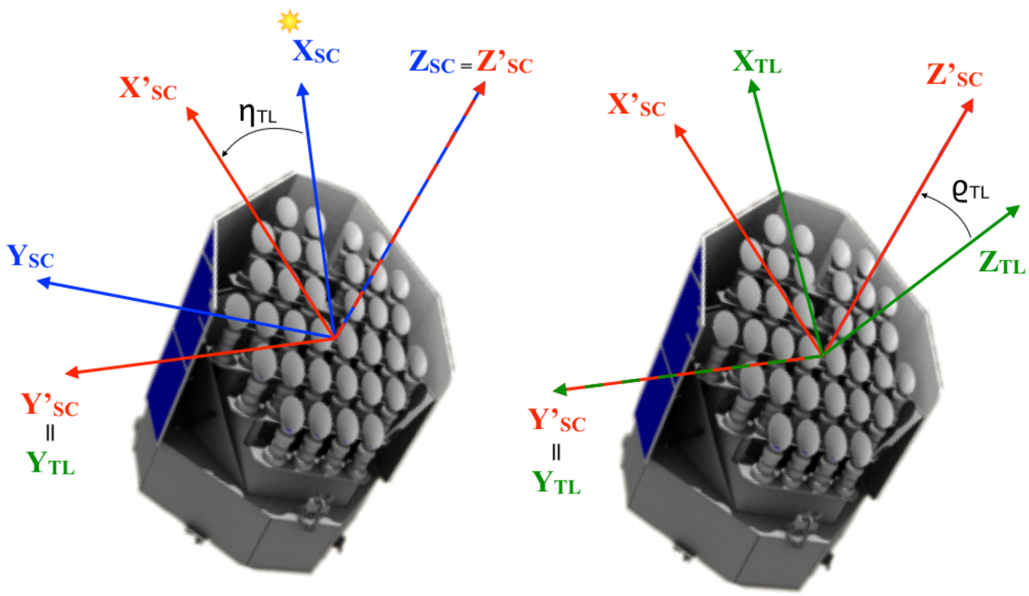

Figure 4: The optical axis $Z_{\rm FP}$ can be obtained from the spacecraft pointing axis $Z_{\rm SC}$ by first rotating the $(X_{\rm SC}, Y_{\rm SC})$ plane around the pointing axis $Z_{\rm SC}$ over an angle $\eta_{\rm SC}$ and then rotating the resulting $Z'_{\rm SC}$ axis over a tilt angle $\rho_{\rm SC}$ around the $Y'_{\rm SC}$ axis.

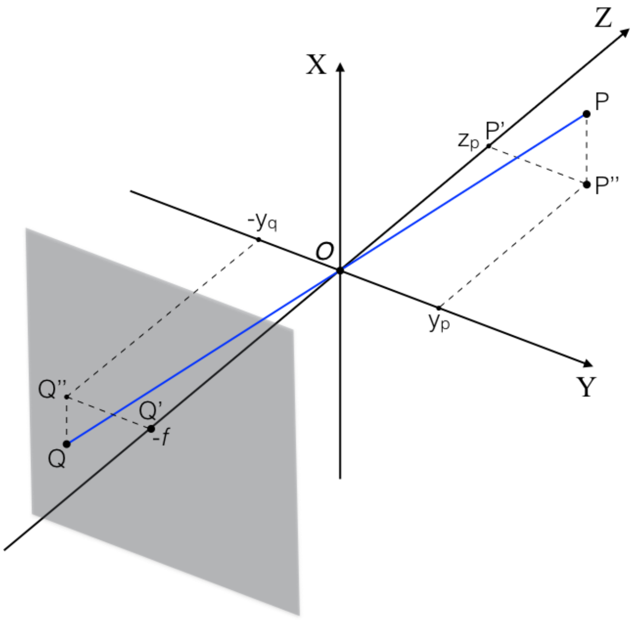

Figure 5: The standard pinhole camera model. The Z axis corresponds to the optical axis of the telescope. The origin of the reference frame is the pinhole O, the focal plane is located behind the pinhole at a distance f where f is the focal length of the camera. Light rays of a star at point P go through the pin hole O and are projected to a point Q on the focal plane.

frame, has coordinates $(x_{\rm TL}, y_{\rm TL}, z_{\rm TL})$ in the telescope reference frame that can be computed using the following transformation:

$$\begin{pmatrix} x_{\rm TL} \\ y_{\rm TL} \\ z_{\rm TL} \end{pmatrix} = \begin{pmatrix} \cos\varrho_{\rm TL} & 0 & -\sin\varrho_{\rm TL} \\ 0 & 1 & 0 \\ \sin\varrho_{\rm TL} & 0 & \cos\varrho_{\rm TL} \end{pmatrix} \begin{pmatrix} \cos\eta_{\rm TL} & -\sin\eta_{\rm TL} & 0 \\ \sin\eta_{\rm TL} & \cos\eta_{\rm TL} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_{\rm SC} \\ y_{\rm SC} \\ z_{\rm SC} \end{pmatrix} \tag{6}$$

where we used the rotation matrices defined in Appendix B. The inverse transformation is:

$$\begin{pmatrix} x_{\rm SC} \\ y_{\rm SC} \\ z_{\rm SC} \end{pmatrix} = \begin{pmatrix} \cos\eta_{\rm TL} & \sin\eta_{\rm TL} & 0 \\ -\sin\eta_{\rm TL} & \cos\eta_{\rm TL} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos\varrho_{\rm TL} & 0 & \sin\varrho_{\rm TL} \\ 0 & 1 & 0 \\ -\sin\varrho_{\rm TL} & 0 & \cos\varrho_{\rm TL} \end{pmatrix} \begin{pmatrix} x_{\rm TL} \\ y_{\rm TL} \\ z_{\rm TL} \end{pmatrix} \tag{7}$$

5 The Focal Plane reference frame

The right-handed cartesian coordinate system $(X_{\rm FP}, Y_{\rm FP}, Z_{\rm FP})$ is associated with the Focal Plane of one camera, and like the telescope reference frame it has its origin exactly in the middle of the 4 CCDs. The $Z_{\rm FP}$ axis coincides with the $Z_{\rm TL}$ optical axis of the telescope which points towards the field-of-view. The focal plane $(X_{\rm FP}, Y_{\rm FP})$ is obtained by rotating the $(X_{\rm TL}, Y_{\rm TL})$ plane over an angle $\gamma_{\rm FP}$ called the focal plane orientation around the $Z_{\rm FP} = Z_{\rm TL}$ axis. The corresponding tranformation is:

$$\begin{pmatrix} x_{\rm FP} \\ y_{\rm FP} \\ z_{\rm FP} \end{pmatrix} = R^t_{\hat{\mathbf{z}}_{\rm FP}}(\gamma_{\rm FP}) \begin{pmatrix} x_{\rm TL} \\ y_{\rm TL} \\ z_{\rm TL} \end{pmatrix} = \begin{pmatrix} \cos\gamma_{\rm FP} & \sin\gamma_{\rm FP} & 0 \\ -\sin\gamma_{\rm FP} & \cos\gamma_{\rm FP} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_{\rm TL} \\ y_{\rm TL} \\ z_{\rm TL} \end{pmatrix} \tag{8}$$

Here $(x_{\rm TL}, y_{\rm TL}, z_{\rm TL})$ are the coordinates of a vector in the telescope reference frame, and $(x_{\rm FP}, y_{\rm FP}, z_{\rm FP})$ are the corresponding coordinates of that same vector in the focal plane reference frame. The inverse transformation is

$$\begin{pmatrix} x_{\rm TL} \\ y_{\rm TL} \\ z_{\rm TL} \end{pmatrix} = \begin{pmatrix} \cos\gamma_{\rm FP} & -\sin\gamma_{\rm FP} & 0 \\ \sin\gamma_{\rm FP} & \cos\gamma_{\rm FP} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_{\rm FP} \\ y_{\rm FP} \\ z_{\rm FP} \end{pmatrix} \tag{9}$$

The projection of the sky on the detector is not done using a simple orthogonal projection on the $(X_{\rm FP}, Y_{\rm FP})$ plane, but is done using a standard pinhole camera model as shown in Fig. 5. The Z axis is the optical axis of the telescope, the (X, Y) plane is parallel to the focal plane, but has its origin in the pinhole. The focal plane is at a distance f behind the pinhole, where f is the focal length of the camera. Light rays of a star at point P go through the pin hole O and are projected to a point Q on the focal plane. In the (Y, Z) plane, the triangles $\widehat{OP'P''}$ and $\widehat{OQ'Q''}$ are similar triangles, so that we have $|P'P''|/|OP'| = |Q'Q''|/|OQ'|$ or

$$\frac{-y_q}{f} = \frac{y_p}{z_p} \tag{10}$$

where the minus sign indicates that the pinhole reverses the image on the focal plane. A similar argument holds in the (X, Z) plane, so that we can write

$$\begin{pmatrix} x_q \\ y_q \end{pmatrix} = -\frac{f}{z_p} \begin{pmatrix} x_p \\ y_p \end{pmatrix} \tag{11}$$

Note that we only need the direction of the point P, i.e. (x_p, y_p, z_p) and (ax_p, ay_p, az_p) lead to the same projected point Q, as long as the normalisation by the z-component is followed in Eq. (11). In practice we therefore compute the coordinates of the star on a unit sphere. Given the equatorial sky coordinates (α, δ), we compute the cartesian coordinates of the star in the equatorial reference frame as

$$\begin{pmatrix} x_{\mathrm{EQ}} \\ y_{\mathrm{EQ}} \\ z_{\mathrm{EQ}} \end{pmatrix} = \begin{pmatrix} \cos\alpha\cos\delta \\ \sin\alpha\cos\delta \\ \sin\delta \end{pmatrix} \tag{12}$$

using Eq. (22), convert to cartesian coordinates $(x_{\mathrm{SC}}, y_{\mathrm{SC}}, z_{\mathrm{SC}})$ in the spacecraft reference frame using Eq. (5), then convert to cartesian coordinates $(x_{\mathrm{TL}}, y_{\mathrm{TL}}, z_{\mathrm{TL}})$ in the telescope reference frame using Eq. (6), then convert to cartesian coordinates $(x_{\mathrm{FP}}, y_{\mathrm{FP}}, z_{\mathrm{FP}})$ in the focal plane reference frame using Eq. (8), and finally do the pinhole projection using Eq. (11).

In the Sections below, when we mention focal plane coordinates $(x_{\mathrm{FP}}, y_{\mathrm{FP}})$ we are always referring to the pinhole-projected coordinates.

6 The CCD reference frame

The CCD coordinate system $(X_{\mathrm{CCD}}, Y_{\mathrm{CCD}})$ is associated with 1 CCD, and has its origin in one of the corners of the CCD. The charge transfer direction during readout of the CCD happens in the negative Y_{CC} direction and the readout of the serial readout register happens in the negative X_{CC} direction. The orientation of the CCD with respect to the $(X_{\mathrm{FP}}, Y_{\mathrm{FP}})$ coordinate system is defined by the angle γ_{CCD}, as shown in Fig. 6.

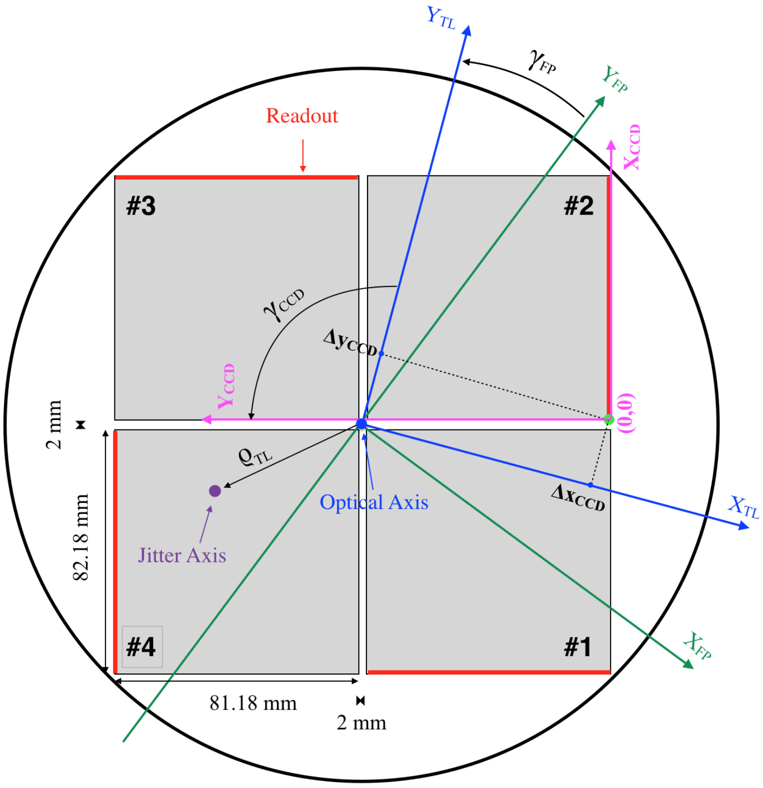

Figure 6: A schematic overview of the focal plane with 4 CCDs. The optical axis $Z_{\rm FP}$ is the blue dot in the middle of the 4 CCDs and points in the positive direction towards the reader. The jitter roll axis $Z_{\rm SC}$ is the purple dot, and also points in the positive direction towards the reader. $\gamma_{\rm CCD}$ is the CCD orientation angle.

When looking into the positive $Z_{\rm FP}$ direction, $\gamma_{\rm CCD}$ increases clockwise. The transformations between the focal plane and the CCD reference frames are given by

$$\begin{pmatrix} x_{\rm FP} \\ y_{\rm FP} \end{pmatrix} = \begin{pmatrix} \cos\gamma_{\rm CCD} & -\sin\gamma_{\rm CCD} \\ \sin\gamma_{\rm CCD} & \cos\gamma_{\rm CCD} \end{pmatrix} \begin{pmatrix} x_{\rm CCD} - \Delta x_{\rm CCD} \\ y_{\rm CCD} - \Delta y_{\rm CCD} \end{pmatrix} \tag{13}$$

and

$$\begin{pmatrix} x_{\rm CCD} \\ y_{\rm CCD} \end{pmatrix} = \begin{pmatrix} \cos\gamma_{\rm CCD} & \sin\gamma_{\rm CCD} \\ -\sin\gamma_{\rm CCD} & \cos\gamma_{\rm CCD} \end{pmatrix} \begin{pmatrix} x_{\rm FP} \\ y_{\rm FP} \end{pmatrix} + \begin{pmatrix} \Delta x_{\rm CCD} \\ \Delta y_{\rm CCD} \end{pmatrix} \tag{14}$$

where $(\Delta x_{\rm CCD}, \Delta y_{\rm CCD})$ are the coordinates of the CCD zeropoint in the $(X_{\rm FP}, Y_{\rm FP})$ frame. An overview of the zero point coordinates and the CCD orientation angle $\gamma_{\rm CCD}$ is given in Table 1.

CCD	$N_{\rm rows}$	$N_{\rm cols}$	$r_{\rm exp}$	$\Delta x_{\rm CCD}$ (mm)	$\Delta y_{\rm CCD}$ (mm)	$\gamma_{\rm CCD}$ (deg)
1	4510	4510	0	-1	82.162	0
2	4510	4510	0	-1	82.162	90
3	4510	4510	0	-1	82.162	180
4	4510	4510	0	-1	82.162	270
1 (F)	4510	4510	2255	-1	82.162	0
2 (F)	4510	4510	2255	-1	82.162	90
3 (F)	4510	4510	2255	-1	82.162	180
4 (F)	4510	4510	2255	-1	82.162	270

Table 1: The first column lists the CCD codes: 1,2,3,4 for a normal camera, 1 (F), 2 (F), 3 (F), 4 (F) for a fast camera. $r_{\rm exp}$ is the first row that is exposed; for the fast camera only rows 2255 until 4510 are exposed. $\Delta x_{\rm CCD}$ and $\Delta y_{\rm CCD}$ are the CCD origin coordinates in the $(X_{\rm FP}, Y_{\rm FP})$ frame. $\gamma_{\rm CCD}$ is the CCD orientation angle.

7 Radial dependency of the PSF

In realistic cases, the PSF is dependent on the angular radial distance θ between the projected center $(x_{\rm FP}, y_{\rm FP})$ of the PSF in the focal plane to the optical axis which is assumed to go through the origin $(0,0)$ of the focal plane. The cosine of this angular distance can be computed using

$$\cos\theta = \frac{\mathbf{v}\cdot\mathbf{z}}{\|\mathbf{v}\|\ \|\mathbf{z}\|} \tag{15}$$

where $\mathbf{v} \equiv (x_{\rm FP}, y_{\rm FP}, -f)$ is the coordinate vector of the PSF in the pinhole reference frame (see Fig. 5), and $\mathbf{z} = (0,0,-1)$ is the unit vector along the optical axis but pointing towards the focal plane, also in the pinhole reference frame. This leads to

$$\cos\theta = \left[1 + \left(\frac{x_{\rm FP}}{f}\right)^2 + \left(\frac{y_{\rm FP}}{f}\right)^2\right]^{-1/2} \tag{16}$$

where $(x_{\rm FP}, y_{\rm FP})$ are the pinhole-projected coordinates of the center of the PSF, and where f is the focal length of the camera.

8 Platform jitter rotation angles

We define the platform yaw φ, the pitch θ, and the roll ψ as rotation angles around respectively the $X_{\rm SC}$, $Y_{\rm SC}$, and $Z_{\rm SC}$ axes, such that the angles increase with a clockwise rotation, when looking along the positive axes. This is illustrated in Fig. 2.

First a roll rotation is done around the $Z_{\rm SC}$ axis, then a pitch rotation is done around the rotated $Y_{\rm SC}$ axis, and finally a yaw rotation is done around the twice-rotated $X_{\rm SC}$ axis. The combined rotation matrix (in the Spacecraft SC reference frame) is given by (cf. Eq. (26)):

$$R(\theta,\varphi,\psi) = R(\theta)R(\phi)R(\psi) \tag{17}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\varphi & -\sin\varphi \\ 0 & \sin\varphi & \cos\varphi \end{pmatrix} \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix} \begin{pmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{18}$$

with the inverse transformation

$$R^{-1}(\theta,\varphi,\psi) = R^t(\psi)R^t(\phi)R^t(\theta) \tag{19}$$

$$= \begin{pmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\varphi & \sin\varphi \\ 0 & -\sin\varphi & \cos\varphi \end{pmatrix} \tag{20}$$

To compute the updated equatorial sky coordinates $(\tilde\alpha_{\rm SC}, \tilde\delta_{\rm SC})$ of the platform Z axis after jitter, we first apply the yaw, jitter, and roll rotations to the current roll axis $\hat{\mathbf{z}}_{\rm SC} = (0,0,1)$ in the spacecraft reference frame:

$$\tilde{\mathbf{z}}_{\rm SC} = R(\theta,\varphi,\psi)\cdot\hat{\mathbf{z}}_{\rm SC} \tag{21}$$

and then use the inverse transformation of Eq. (5) and Eq. 23 to compute the updated $(\tilde\alpha_{\rm SC}, \tilde\delta_{\rm SC})$.

9 Telescope drift rotation angles

TBW

Acknowledgements

We thank Denis Grießbach of DLR for helpful comments on an early version of the document.

A Spherical Coordinate Transformation

We use the following notation for the standard 3D spherical coordinate transformation:

$$f : \mathbb{R}^+ \times [0,\pi] \times [0,2\pi] \to \mathbb{R}^3 : \begin{pmatrix} r \\ \theta \\ \phi \end{pmatrix} \mapsto \begin{pmatrix} x \\ y \\ z \end{pmatrix} \equiv \begin{pmatrix} r\sin\theta\cos\phi \\ r\sin\theta\sin\phi \\ r\cos\theta \end{pmatrix} \tag{22}$$

and its inverse:

$$f^{-1} : \mathbb{R}^3 \to \mathbb{R}^+ \times [0,\pi] \times [0,2\pi] : \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto \begin{pmatrix} r \\ \theta \\ \phi \end{pmatrix} \equiv \begin{pmatrix} \sqrt{x^2+y^2+z^2} \\ \arccos\left(z/\sqrt{x^2+y^2+z^2}\right) \\ \arctan(y/x) \end{pmatrix} \tag{23}$$

B Rotation Matrix

We consider a vector $\mathbf{v} = v_x\mathbf{e}_x + v_y\mathbf{e}_y + v_z\mathbf{e}_z$ that is rotated by an angle θ around an axis determined by the unit vector $\hat{\mathbf{u}} = u_x\mathbf{e}_x + u_y\mathbf{e}_y + u_z\mathbf{e}_z$, to give a vector $\mathbf{v}' = v'_x\mathbf{e}_x + v'_y\mathbf{e}_y + v'_z\mathbf{e}_z$. The components of the two vectors are related by:

$$\begin{pmatrix} v'_x \\ v'_y \\ v'_z \end{pmatrix} = R_{\mathbf{u}}(\theta) \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} \tag{24}$$

and

$$\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = R^{-1}_{\mathbf{u}}(\theta) \begin{pmatrix} v'_x \\ v'_y \\ v'_z \end{pmatrix} = R^t_{\mathbf{u}}(\theta) \begin{pmatrix} v'_x \\ v'_y \\ v'_z \end{pmatrix} \tag{25}$$

where $R_{\mathbf{u}}(\theta)$ is the rotation matrix given by

$$R_{\mathbf{u}}(\theta) = \begin{pmatrix} \cos\theta + u_x^2(1-\cos\theta) & u_xu_y(1-\cos\theta) - u_z\sin\theta & u_xu_z(1-\cos\theta) + u_y\sin\theta \\ u_yu_x(1-\cos\theta) + u_z\sin\theta & \cos\theta + u_y^2(1-\cos\theta) & u_yu_z(1-\cos\theta) - u_x\sin\theta \\ u_zu_x(1-\cos\theta) - u_y\sin\theta & u_zu_y(1-\cos\theta) + u_x\sin\theta & \cos\theta + u_z^2(1-\cos\theta) \end{pmatrix} \tag{26}$$

and where $R^t_{\mathbf{u}}(\theta)$ denotes the transposed matrix. Looking into the positive direction of $\mathbf{u}$ the rotation angle θ increases clockwise (right-hand rule).

In the rotated reference frame $(\mathbf{e}_{x'}, \mathbf{e}_{y'}, \mathbf{e}_{z'}) \equiv (R_{\mathbf{u}}(\theta)\mathbf{e}_x,\ R_{\mathbf{u}}(\theta)\mathbf{e}_y,\ R_{\mathbf{u}}(\theta)\mathbf{e}_z)$, we can look at the same vectors $\mathbf{v} = v_{x'}\mathbf{e}_{x'} + v_{y'}\mathbf{e}_{y'} + v_{z'}\mathbf{e}_{z'}$ and $\mathbf{v}' = v'_{x'}\mathbf{e}_{x'} + v'_{y'}\mathbf{e}_{y'} + v'_{z'}\mathbf{e}_{z'}$. In this case we have

$$\begin{pmatrix} v_{x'} \\ v_{y'} \\ v_{z'} \end{pmatrix} = R^t_{\mathbf{u}}(\theta) \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} \tag{27}$$

and

$$\begin{pmatrix} v'_{x'} \\ v'_{y'} \\ v'_{z'} \end{pmatrix} = R^t_{\mathbf{u}}(\theta) \begin{pmatrix} v'_x \\ v'_y \\ v'_z \end{pmatrix} = \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} \tag{28}$$

As a final remark, we observe that

$$R_{-\mathbf{u}}(-\theta) = R_{\mathbf{u}}(\theta). \tag{29}$$